CHL8010 Proposal

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Abstract

This document outlines the proposal for conducting a literature review to identify research gaps, formulate clear research questions, and specify relevant data tables and variables from the chosen dataset. It includes an initial plan for data wrangling and statistical analysis, ensuring thorough familiarization with the dataset, alongside a comprehensive project management plan.

Keywords

Literature Review, Data Wrangling, Survival Analysis, Longitudinal Data, Asthma, Depression, Chronic Obstructive Pulmonary Disease (COPD)

1 Introduction and Literature Review

This section synthesizes findings from high-impact academic sources to identify a specific research gap in the current literature regarding respiratory diseases and mental health comorbidities.

1.1 Literature Context

Chronic obstructive pulmonary disease (COPD) represents a substantial and growing public health burden in Canada and worldwide [1]. According to the statistics reported from the Canadian Chronic Disease Surveillance System in 2018, 3.8 million Canadians over the age of one suffer from asthma, and 2.0 million suffer from COPD, which could impact their personal ability to breathe [4]. Beyond the physiological impact on respiratory function, both conditions are frequently comorbid with psychiatric disorders.

Extensive research has documented a significantly higher burden of depression among these populations, such that respondents with COPD or asthma usually report worse overall mental health status and a greater risk of depressive symptoms [2]. For instance, a cross-sectional analysis of the Cooper Center Longitudinal Study found that adults with asthma showed that an increased prevalence of both current and lifetime depression is independent of disease severity [5], though some evidence suggest that the likelihood of suffering from mental health disorders is likely to increase as the severity of asthma [3]. Similarly, population-based cohort studies have shown that symptomatic asthma in youth increases the likelihood of a wide range of mental health problems [3]. The cohort and nested case control study conducted by Walters et al. [6] also indicates that patients with asthma have a higher rate of depression compared to patients without asthma.

1.2 Research Gap

Although a large amount of research shows that people with asthma and COPD are more likely to experience depressive symptoms, there is still a significant knowledge gap regarding the varying risk profiles of different respiratory phenotypes. Few studies have investigated the incidence rate of depression by distinctly comparing asthma to other conditions within the classification of COPD. It remains unclear whether the trajectories of incident depression in patients with asthma is different to those in patients diagnosed with other non-asthmatic COPD. Addressing this gap requires shifting from cross-sectional prevalence estimates to longitudinal incidence tracking using standardized diagnostic codes in real-world clinical settings.

2 Research Question

Drawing upon the gaps identified in the literature, the following research questions were formulated. The variables selected for this study were explicitly chosen to address these specific clinical unknowns. To address this gap, this project uses real-world primary care electronic medical record data from the Canadian Primary Care Sentinel Surveillance Network (CPCSSN).

> **Primary Research Question:** *How does the incidence of new-onset mental health conditions (specifically depression) differ between patients with asthma compared to those with non-asthmatic chronic obstructive pulmonary disease?*

The purpose of this study is to examine and compare the true incidence rates of depression in these two cohorts. To precisely isolate incidence and reduce the confounding effect of high baseline psychiatric comorbidities, the study design will exclude all patients with any documented mental health records prior to their initial (index) diagnosis date of asthma or COPD.

3 Data Wrangling and Analysis Plan

The primary analytical dataset will be constructed using the `EncounterDiagnosis`, `Lab`, `Patient`, and `Family` tables as the foundational source, as it contains patient-level diagnostic records with ICD-9 codes and encounter dates. The data wrangling process will proceed in a structured, multi-stage pipeline designed to ensure data integrity, temporal validity, and analytic readiness (See Figure 1).

3.1 Data Cleaning and Standardization

Systematic data validation and cleaning is done first to ensure structural consistency.

3.1.1 **Duplicate Detection:** `EncounterDiagnosis_ID` is used as a unique key. Repeated entries will be removed.

3.1.2 **Dealing with Missing Values:** All missing values will be standardized (e.g., converting blanks, `NA`, `null`, or placeholder values to consistent `NaN`). Missingness mechanisms will be analyzed and documented.

3.1.3 **Categorical Variable Standardization:** All categorical columns (e.g., diagnosis codes, categorical attributes) will be converted to standardized uppercase string format to eliminate case-sensitivity inconsistencies.

3.1.4 **Date Variable Processing:** All date columns will be converted to proper datetime format. Invalid or malformed dates (e.g., impossible calendar dates or future dates beyond study end) will be flagged and reported. Records with invalid dates will be reviewed and excluded if necessary.

3.1.5 **Numeric Validation:** Numeric columns will be converted to appropriate numeric types. Negative values in biologically implausible fields (e.g., age, lab values) will be removed or corrected after validation.

3.2 Cohort Identification: COPD Population

Using cleaned diagnostic records, patients with COPD and allied conditions (ICD-9 codes 490 – 496) will be filtered. ICD-9 codes will be extracted and filtered using standardized string matching to capture all eligible COPD-related diagnoses.

3.3 Index Date Construction

For each identified COPD patient, an index date will be constructed: The index date will be defined as the earliest encounter date on which a COPD-related ICD-9 code (490 – 496) was recorded. This index date represents the start of risk follow-up and ensures a consistent temporal anchor for outcome assessment.

3.4 Exclusion of Prevalent Depression Cases

This step ensures that the resulting cohort represents patients at risk for incident depression following COPD diagnosis. To study incidence rather than prevalence, patients with documented depression prior to the index date will be excluded.

3.4.1 Mental health conditions are defined using ICD-9: 290 - 319. Depression is defined using ICD-9 codes: 296.2x, 296.3x, 300.4, 311, 309.0.

3.4.2 For each patient, all diagnosis records prior to the index date will be examined. Patients with any depression diagnosed before index will be excluded.

3.5 Population Definition

The study population contains patients who are confirmed COPD, valid index date, at risk of depression, and have valid demographic and follow-up information.

3.6 Patient-Level Feature Engineering

After cohort construction, encounter-level data will be converted into a patient-level analytical dataset (See Table 1) , which includes possible tables:

3.6.1 **Patient Demorgraphics and Diagnoses:** Patient ID, demographic variables (age at index, sex, etc.), index date, diagnosis history prior to index. Diagnosis records defining the primary outcome (Depression) and the specific exposure cohorts (Asthma and Non-Asthma COPD) will be identified using ICD-9 codes.

3.6.2 **Number of Comorbidities:** The number of unique records prior to index.

3.6.3 **Biomarkers and Laboratory Values:** From `lab` tables and aligned relative to index date (e.g., most recent lab value within a defined baseline window).

3.6.4 **Family History Indicators:** From `Family` tables, indicators of family history can be derived by evaluating diagnosis text in conjunction with patient relationship strings.

All time-dependent variables will be carefully restricted to baseline (pre-index) periods to prevent information leakage.

Table 1: Predictors and Outcomes for Analysis

Predictor	Table	Variable Name	Description
Outcome			
Depression	`Encounter Diagnosis`	`DiagnosisCode_calc`	ICD-9 codes for depressive disorders.
Exposure			
Asthma	`Encounter Diagnosis`	`DiagnosisCode_calc`	ICD-9 codes for asthma diagnosis.
Non-Asthma COPD	`Encounter Diagnosis`	`DiagnosisCode_calc`	ICD-9 codes for all COPD (excl. asthma) diagnosis.
Covariates			
Age	`Patient`	`BirthYear`	Initial Diagnosis Year - BirthYear.
Sex	`Patient`	`Sex`	Patient biological sex.
BMI	`Lab`	`TestResult_calc`	Body Mass Index from lab results.
HbA1c	`Lab`	`TestResult_calc`	Lab result for glycated hemoglobin percentage.
Glucose	`Lab`	`TestResult_calc`	Lab result for blood glucose levels.
Number of Comorbidities	`Encounter Diagnosis`	`DiagnosisCode_calc`	Count of unique records prior to index.
Family History	`Family`	`DiagnosisText_orig` & `Relationship_orig`	Extracted by matching diagnosis text with relationship strings this is patient's family member relationship and family's diagnosis.

3.7 Data Integration

To enrich the analytical dataset, relevant tables will be merged at the patient level: `Patient`, `EnounterDiagnosis` (3.6.1, 3.6.2), `lab` (3.6.3), and additional clinical or family history tables (e.g., `Family`) as needed (3.6.4). All joins will be validated to ensure one-to-one or appropriately aggregated merges. In cases of multiple baseline values (e.g., multiple labs), clinically justified selection rules (e.g., most recent prior to index) will be applied.

3.8 Final Dataset

The final dataset will be structured at the patient level with:

- Binary outcome variable (incident depression post-index)
- Primary exposure (COPD status)
- Adjusted covariates
- Valid temporal ordering (baseline → index → follow-up)

This structured pipeline ensures reproducibility, temporal validity, and analytic transparency in preparation for regression modeling.

3.9 Analysis Plan

3.9.1 Descriptive Survival Analyses and Modeling

Participants will be divided into two cohorts based on the primary exposure (asthma vs. non-asthma COPD). Kaplan–Meier curves and cumulative incidence curves for each cohort will be used to visualize the survival experience over time. To compare the survival distributions between the two cohorts, a two-sided log-rank test will be conducted.

A multivariable Cox proportional hazards model will be fitted to evaluate the association between the exposure and the outcome while adjusting for relevant covariates. The model will include all available variables, such as asthma status, family history, education level, employment status, age, and other demographic or clinical characteristics. Adjusted hazard ratios and corresponding confidence intervals will be reported to quantify the effects of these variables on the time-to-event outcome.

3.9.2 Software and Packages

All analyses will be conducted using Python. The following packages will be used to support data processing, statistical modeling, and visualization. The `pandas` and `numpy` packages will be used for data cleaning, data manipulation, and numerical operations required during the data wrangling process. Survival analyses will be performed using the `lifelines` package. Figures such as Kaplan–Meier survival curves and cumulative incidence plots will be generated using `matplotlib` to ensure clear and reproducible visualization of the results.

4 Project Management Plan

The whole project is split into two phases. Phase One focuses on building `main.py`. To ensure that everyone understands the project workflows, all group members participate in brainstorming sessions and frequent group discussions. Yiting is responsible for the modular design after brainstorming. Shirling is responsible for module testing using CPCSSN data. Min is responsible for modular flexibility testing. A mock peer evaluation is conducted at the end of Phase One.

Phase Two focuses on poster design and result visualization. After the project interview on April 1st, 2026, group members work on result wrap-up and poster design. Min is responsible for the abstract and introduction, as well as poster design. Shirling is responsible for the literature review and the conclusion and impact sections. Yiting is responsible for the methodology and key findings. All group members participate in the peer evaluation after the poster is submitted.

The detailed project management timeline can be found in the Gantt chart, `Guide_ProjectManagement_Group1.xlsx`.

4.1 Timeline and GitHub Integration

Consistent with the submitted `Guide_ProjectManagement_Group1.xlsx`, GitHub history will show frequent, meaningful plan updates and progress reflections.

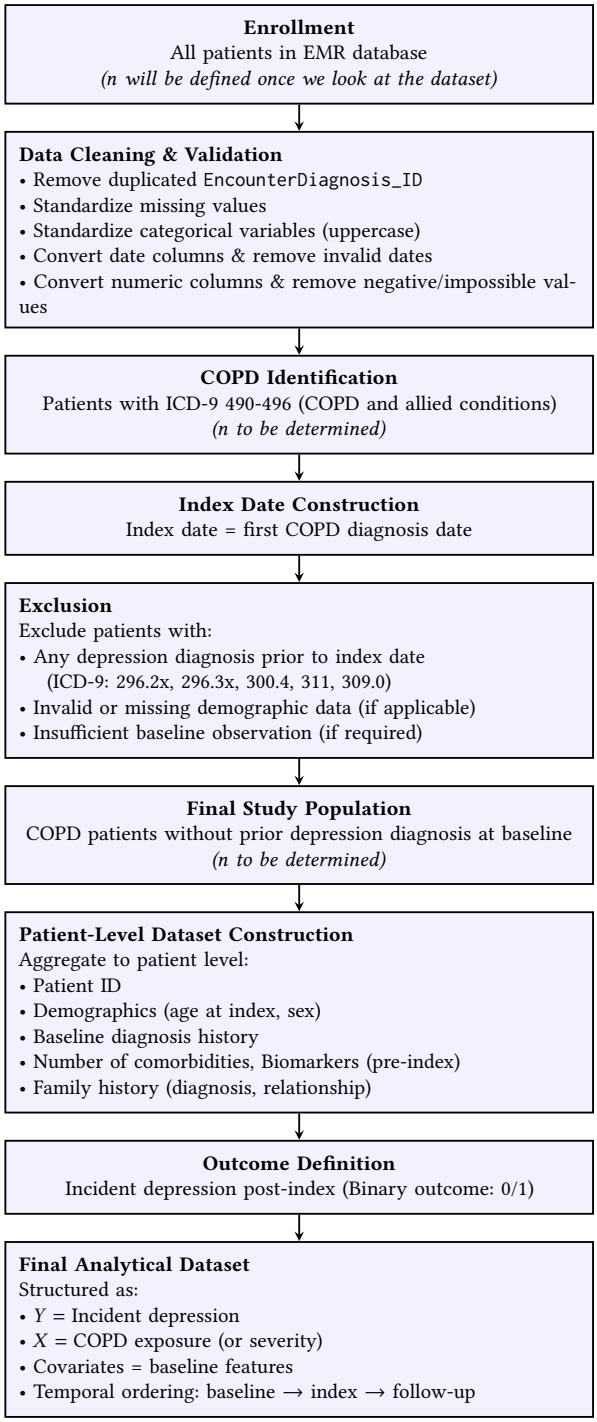

Figure 1: Data Wrangling, Cohort Selection, and Analytical Dataset Construction Pipeline

References

[1] M. DiBonaventura, P. R. daCosta, J. Su, et al. 2012. The burden of chronic obstructive pulmonary disease among employed adults. *International Journal of Chronic Obstructive Pulmonary Disease* 7 (2012), 211–219. doi:10.2147/COPD.S29280

[2] M. D. Eisner, E. H. Yelin, L. Trupin, and P. D. Blanc. 2002. The influence of chronic respiratory conditions on health status and work disability. *American Journal of Public Health* 92, 9 (2002), 1506–1513. doi:10.2105/ajph.92.9.1506

[3] R. D. Goodwin, M. Robinson, P. D. Sly, et al. 2013. Severity and persistence of asthma and mental health: a birth cohort study. *Psychological Medicine* 43, 6 (2013), 1313–1322. doi:10.1017/S0033291712001754

[4] Public Health Agency of Canada. 2018. Asthma and chronic obstructive pulmonary disease (COPD) in Canada, 2018. Government of Canada. https://www.canada.ca/en/public-health/services/publications/diseases-conditions/asthma-chronic-obstructive-pulmonary-disease-canada-2018.html

[5] Timothy D. Trojan, David A. Khan, Laura F. DeFina, Oviea Akpotaire, Renee D. Goodwin, and E. Sherwood Brown. 2014. Asthma and depression: the Cooper Center Longitudinal Study. *Annals of Allergy, Asthma & Immunology* 112, 5 (2014), 432–436. doi:10.1016/j.anai.2014.02.015

[6] P. Walters, P. Schofield, L. Howard, M. Ashworth, and A. Tylee. 2011. The relationship between asthma and depression in primary care patients: A historical cohort and nested case control study. *PLoS ONE* 6, 6 (2011), e20750. doi:10.1371/journal.pone.0020750